

COURSE SPECIFICATION
OBJECT-ORIENTED PROGRAMMING (2101623)

1. Course name and code: Object-Oriented Programming -2101623

2. Total no. of credits: 3

Theoretical: 2

Practical: 1

Self-study: 5

3. Instructor's or course coordinator's name

MSc. Nguyễn Thị Hoàng Khánh

MSc. Châu Thị Bảo Hà

MSc. Nguyễn Văn Thắng

MSc. Phạm Thanh Hùng

MSc. Trần Thị Anh Thi

MSc. Đặng Văn Thuận

4. Learning materials

Main textbooks

[1] Scott Oaks, Java Performance, 2nd ed. , O'Reilly Media, Inc., 2020. [FIT_SE_001]

Other supplemental materials

[1] Cay S. Horstmann, Core Java. Volume I, Fundamentals features , Boston : Prentice Hall, 2016. [100287779]

[2] C. Thomas Wu , An introduction to object-oriented programming with Java, Boston : McGraw Hill Higher Education, 2010.[100281471]

[3] Java Reference <https://www.codecademy.com/learn/learn-java>

5. Specific course information

a. Objectives for the course: The course provides students with:

- Understand the benefits of object-oriented software development.
- Understand basic concepts and terminology in object-oriented programming
- Understand the role of UML in analysis and design, meaning of common schemas
- Apply UML in requirements gathering, analysis and design
- Realize designs in programming languages
- Understand the impact of design so that the application can adapt to changing user requirements.

b. Brief description of the content of the course

- The course provides students with object-oriented approach and application in programming.
- Help students experience through the analysis and design stages using UML through case studies.
- Students learn how to use Java to implement the program.
- Tools and learning materials: computer with Java programming support tools installed

c. Others: At the request of the lecturer Prerequisites or co-requisites

- Prerequisites: Pre-course Programming Techniques (2101405)
- Co-requisites: none

d. Other requirements

None

6. Specific goals for the course

a. Course learning outcomes

Upon successful completion of this course, students should be able to:

No	Course Learning Outcomes (CLOs)	ELOs	Bloom level
1	Explain the advantages of the object-oriented programming (Abstraction, Encapsulation, Inheritance, Polymorphism)	b-PI3	4
2	Identify the task that needs to be solved and the associated constraints based on the description of the problem.	a-PI1	2
3	Present the necessary steps to solve the required problem.	c-PI2	2
4	Implement the program according to the requirements (constraints) of the problem by using the object-oriented programming	b-PI2	3
5	Operate test cases to test the functions of the object according to the equivalent boundary/class method	b-PI2	3

b. Mapping the course learning outcomes and the expected learning outcomes

ELOs CLOs	a	b	c	d	e	f
1		I				
2			I			
3		I				
4			I			
5			I			

c. Mapping CLOs, teaching and learning approach

CLOs	Teaching and learning approach					
	Explicit Teaching	Discussion	Case Studies	Practical activities, experiments	Brainstorming	Project-based learning(PBL)
1	X		X			X
2	X	X			X	
3	X	X	X	X		
4	X		X	X	X	
5	X	X		X	X	

d. Mapping CLOs and assessment methods

CLOs	Assessment methods		
	Performance test	Case studies	Project
1		X	
2			X
3	X		
4	X	X	
5		X	X

7. Contents and plan

No.	Learning Contents	Lesson Learning Outcomes [LLO]	Hours Lecture / Practice	CLOs	Teaching and learning methods		Self-study content and instructions
					What lecture does	What students do	
1	CHAPTER 1: OVERVIEW OF THE OBJECT-ORIENTED APPROACH 1.1 The approach of traditional programming 1.2 Object-oriented approach 1.3 Compare the difference between the two approaches 1.4 Development trend of object-oriented programming	LLO 1: Understanding the Traditional Programming Approach LLO 2: Understand and present object-oriented approach LLO 3: Compare the difference between the two approaches LLO 4: Understanding the development trend	3/0	1,2	Explicit teaching; and raises some situational questions; makes model exercises	Take notes; ask or answer relevant questions;	Read in learning materials: [1]: Chapter 1. [2]: Read more references

		of object-oriented programming					
2	CHAPTER 2: BASIC CONCEPTS OF OBJECT-ORIENTED PROGRAMMING 2.1 Object concept 2.2 Object definition and illustration 2.3 Comparing Classes and Structures 2.4 Describe the Private and Public components of the Class 2.5 Defining Class functions 2.6 Methods of using objects and member functions of Class 2.7 The popular object-oriented programming languages today	LLO 6: Understand the basic concepts, describe the components of object-oriented programming LLO 7: Compare Classes and Structures LLO 8: Methods of using objects and member functions of Class LLO 9: The most popular object-oriented programming languages today	8/3	3,4	Explicit teaching; and raises some situational questions; makes model exercises; instruction in serving as model on the eclipse IDE ; observe and guide students to practice	Take notes; ask or answer relevant questions; do exercises; use the java software to code	Read in learning materials: [1]: chapter 2, [2]: website: https://www.codecademy.com/learn/learn-java Read more references
3	CHAPTER 3: INTRODUCTION TO JAVA	LLO 10: Students can install the program	6/3	4,5	Explicit teaching; and raises some situational	Records; ask or answer relevant questions; do homework;	Read in learning materials: [1]: chapter 3,

	3.1 Development History of Java 3.2 Program architecture built on Java 3.3 Basic Data Structures on Java 3.4 Command Structures on Java 3.5 Case Study I(simple programs & language)	according to the requirements (constraints) of the problem by using the object-oriented programming LLO 11: Students can generate test cases to test the functions of the object according to the equivalent boundary/class method			questions; makes model exercises; instruction in serving as model on the ; observe and guide students to practice	Use java software to solve the exercises and submit on LMS.	Read more references
4	CHAPTER 4: Inheritance and Polymorphism 4.1 Inheritance 4.2 Interface 4.3 Abstract Classes 4.4 Polymorphism 4.5 Schemas in UML 4.6 Exception handling 4.7 Case Study II Object	LLO 12: Students explain the advantages of the object-oriented programming (Abstraction, Encapsulation, Inheritance, Polymorphism)	8/15	1,4,5	Explicit teaching; and raises some situational questions; makes model exercises; instruction in serving as model on the Violet UML Editor tool; observe	Take notes; Ask or answer relevant questions; Do exercises; Practices; Create the lab report	Read in learning materials: [1]: chapter 4, [2]: more references

	Oriented Programs	LL13: Students can install the program according to the requirements (constraints) of the problem by using the object-oriented programming			and guide students to practice		
5	CHAPTER 5: COLLECTIONS 5.1 Concept of Collection 5.2 Comparing Collections and Arrays 5.3 Java Collection Classes 5.4 Applications of Collections in Programming	LLO 14: Students can install the program according to the requirements (constraints) of the problem by using the object-oriented programming LLO 15: Students can generate test cases to test the functions of the object according	5/9	4,5	Explicit teaching; Raises some situational questions; Give students instruction to do the exercises; Observe and guide students to practice	Take notes; Ask or answer relevant questions; Do exercises; Practices; Create the lab report and submit on LMS.	Read in learning materials: [1]: chapter 5, [2] Read more references

		to the equivalent boundary/ class method					
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8. Assessment

a. Description of the assessment methods used based on the Course's Learning Outcomes

(CLOs)	When data collected	Assessment Methods	Assessment Tools	Target of students achieving CLOs (%)
1	Regular test (<i>Writing test</i>)	Continuous	Rubrics	60
2	Project (<i>Presentation</i>)	Continuous	Formative Assessment	65
3	Project (<i>Presentation</i>)	Continuous	Formative Assessment	70
4	Regular test (<i>Practical test</i>)	Continuous	Formative Assessment	75
	Mid-term (<i>Practical test</i>)	Continuous	Formative Assessment	65
	Final-term (<i>Practical test</i>)	Continuous	Rubrics	70
5	Regular test (<i>Practical test</i>)	Summative	Rubrics	70
	Final-term (<i>Practical test</i>)	Summative	Rubrics	60

b. Assessments weighting

Assessment Methods		Weight %
Theory	Continuous (<i>Writing test</i>)	5
	Continuous (<i>Practical test</i>)	5
	Continuous (<i>Practical test</i>)	5
	Summative (<i>Practical test</i>)	5
Practice	Mid-term test (<i>Practical test</i>)	30
	Summative (<i>Practical test</i>)	50

c. Assessment Scale: Credit System

Lecturer: MSc. Dang Van Thuan

Head of department: Dr. Nguyen Thi Hanh